Comp90042 Natural Language Processing

Workshop (Week3 | 2026s1)

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Agenda

1. Discussion (~45 mins)
   - **Text Classification**
   - **N-gram Language Model**

2. Programming (~10 mins)

3	workshop-03.pdf	03-classification.ipynb 04-ngram.ipynb

Discussion : Text classification

1. What is **text classification**? Give some examples.

   (a) Why is text classification generally a difficult problem? What are some hurdles that need to be overcome?

   (b) Consider some (supervised) text classification problem, and discuss whether the following (supervised) machine learning models would be suitable:

      i. k-Nearest Neighbour using Euclidean distance
      ii. k-Nearest Neighbour using Cosine similarity
      iii. Decision Trees using Information Gain
      iv. Naive Bayes
      v. Logistic Regression
      vi. Support Vector Machines

Text Classification

What is Text Classification?

- Is the task of classifying text documents into different labels.

Input

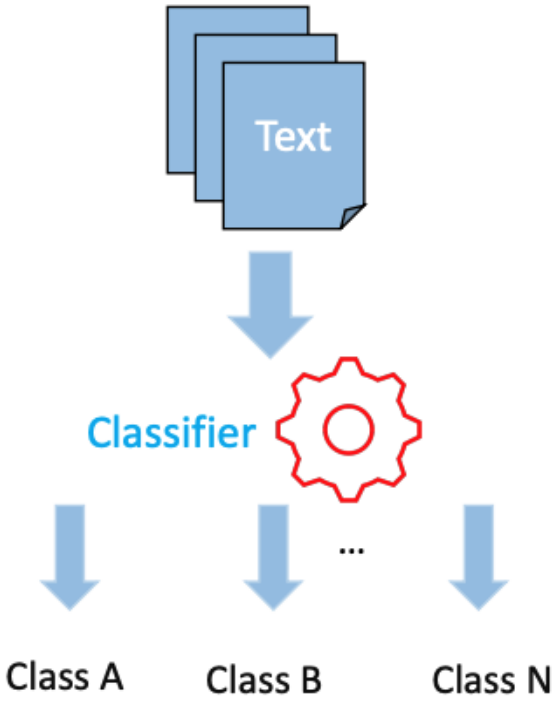
- a document d
- a fixed output set of classes (labels) $C = \{c_1, c_2, \ldots, c_j\}$

(ML run) a training set of $\mathbf{m}$ labeled documents $(d_1, c_1), \ldots, (d_m, c_m)$

Output

- a predicted class $c \in C$

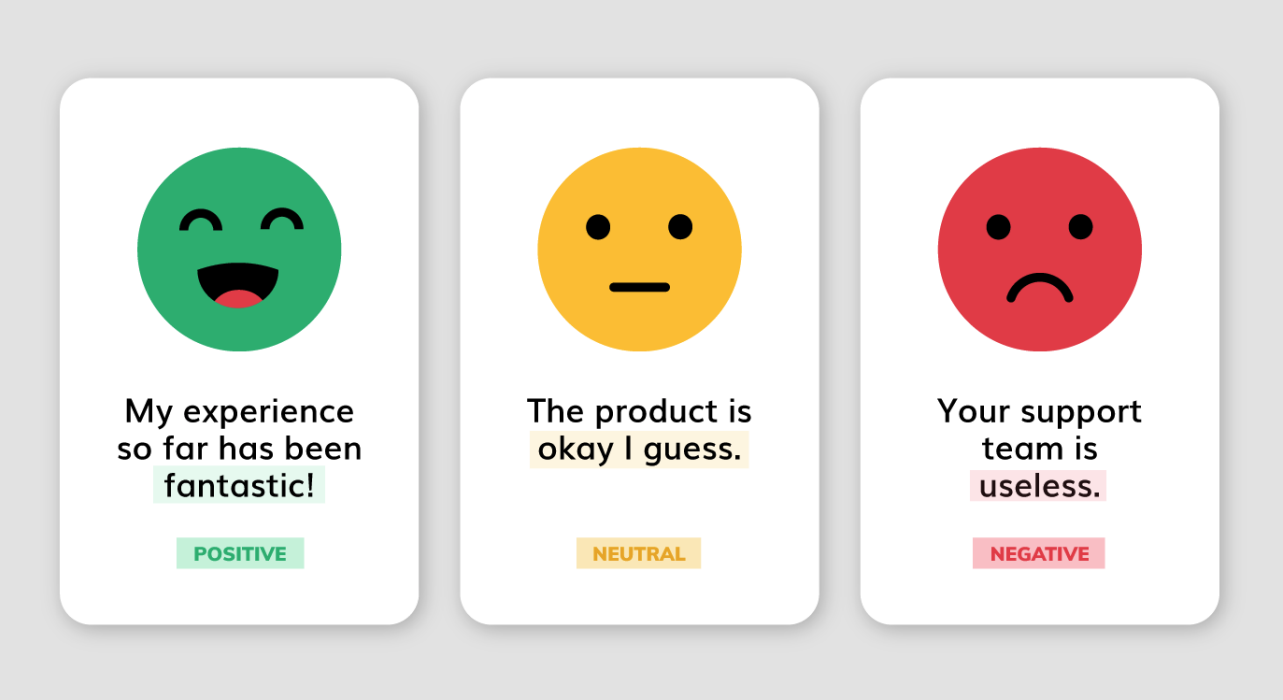
(ML run) a learned classifier $\gamma: d \rightarrow c$



Text Classification

Text Classification Examples

- Sentiment analysis
- Spam detection
- Topic classification
- Authorship identification
- Native-language identification
- Automatic fact-checking
- …



Text Classification - Challenge

Why is text classification generally a difficult problem?

- The main issue is in terms of **document representation**.
- *how do we identify **features** of the **document** which help us to distinguish between the various classes?*

- **Source** of document features: **tokens (words)** in the **document**
  - **feature selection** is often important
  - Single words: inadequate at modelling meaningful information
  - Multi-word (e.g. bi-grams): sparse data problem

Text Classification – Word Representation

How do we represent the meaning of a word?

- ***Count-based word representation :*** *(e.g.) One-hot vectors, Bag of Words, Term Frequency-Inverse Document Frequency (TF-IDF)*
- ***Prediction(Neural)-based word representation :*** *(e.g.) Word2Vec, GloVe*

week5

One-hot vectors (encoding)

- regard words as discrete symbols:

  motel = [0 0 0 0 0 0 0 0 0 0 1 0 0 0 0]

  hotel = [0 0 0 0 0 1 0 0 0 0 0 0 0 0 0]

issues:

➔ *no natural notion of similarity*

➔ *number of words in vocabulary (inefficiency)*

Ref.: NLP lecture note (by Dr Caren Han)

Text Classification – Word Representation

How do we represent the meaning of a word?

Bag of Words (BoW)

- a representation of text that describes **the occurrence of words** within a document.
- The intuition is that documents are similar if they have similar content.

S1= I **love** you but you **hate** me

S2= I **hate** you but you **love** me

issues:

➔ *Discarding word order*

➔ *ignores the context*

➔ *ignores meaning of words in the document (semantics).*

Ref.: NLP lecture note (by Dr Caren Han)

Text Classification – ML models

Consider some (supervised) text classification problem, which of the following ML models would be suitable?

Select one classifier and explain it to your partner. Each of you should choose a different classifier.

- k-Nearest Neighbour (kNN) using Euclidean distance
- k-Nearest Neighbour (kNN) using Cosine similarity
- Decision Trees using Information Gain
- Naive Bayes
- Logistic Regression
- Support Vector Machines

Prerequisites

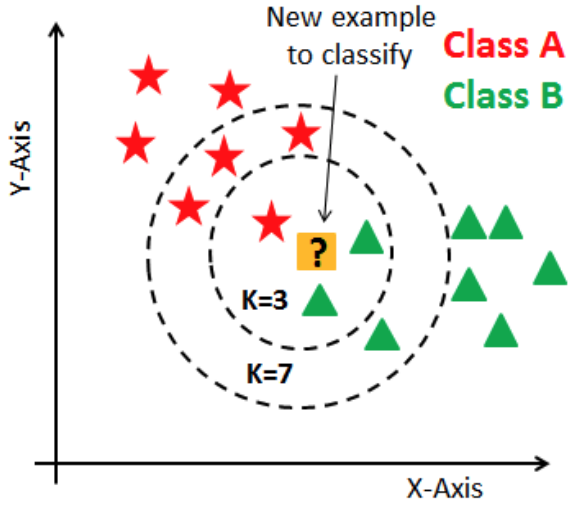
- Machine learning basics (COMP30027, COMP90049, COMP90051)
  - Modules → Welcome → Machine Learning and Linguistics Readings

https://canvas.lms.unimelb.edu.au/courses/234957/pages/machine-learning-and-linguistics-readings?module_item_id=7329045

https://www.youtube.com/watch?v=E0Hmnixke2g

k-Nearest Neighbour (kNN)

- KNN: Classify based on **majority class of k-nearest** training examples in feature space.

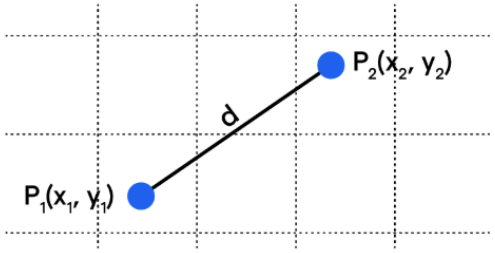


- Distance metric
- Choosing k (e.g. k =3 or ?)

(+) easy to implement, few hyperparameters

(-) not good for scale and high-dimensional data, overfitting

- k-Nearest Neighbour using **Euclidean distance**
  - Often this is a ***bad idea***
  - tends to classify documents **based upon their length**
  - which is usually not a distinguishing characteristic for text classification problems.

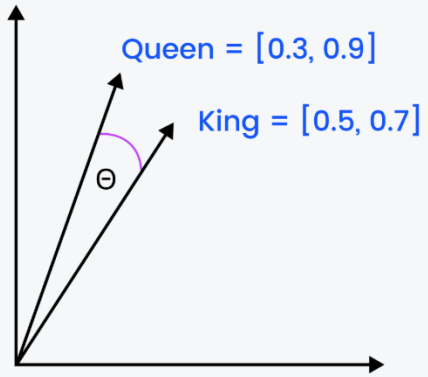


Euclidean Distance (d) = $\sqrt{(x_2 - y_1)^2 + (y_2 - y_1)^2}$

k-Nearest Neighbour (kNN)

- k-Nearest Neighbour using **Cosine similarity**
  - Usually better than using Euclidean distance
  - However, **kNN** suffers from **high-dimensionality problems**
  - our feature set based upon the presence of (all) words usually ***isn't suitable*** for this model.

- **Cosine similarity**
  - measures the similarity between two vectors of an inner (dot) product space.
  - the cosine of the angle between two vectors.



$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}}$$

$$\text{Cos(Queen, King)} = \frac{(0.3*0.5)+(0.9*0.7)}{\sqrt{0.3^2+0.9^2} * \sqrt{0.5^2+0.7^2}}$$

$$= \frac{0.15+0.63}{\sqrt{0.9^2} * \sqrt{0.74}}$$

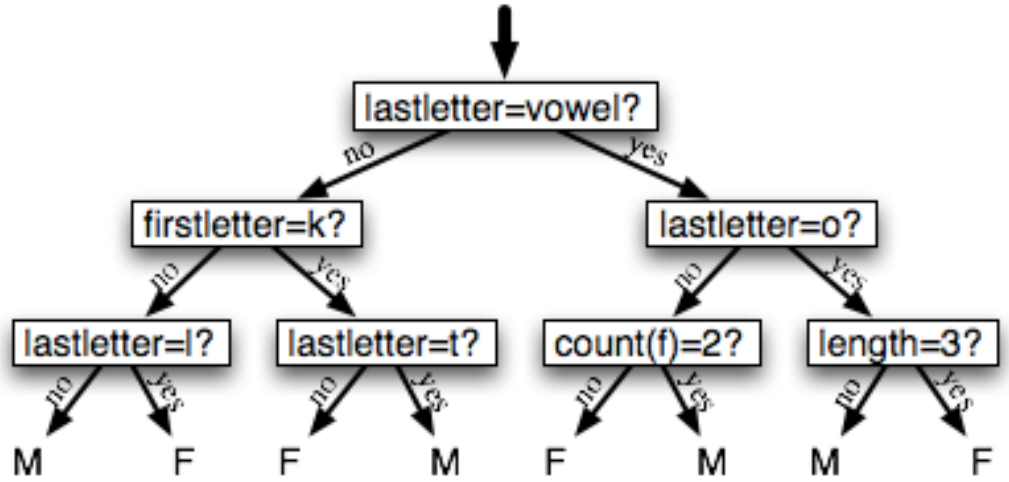
$$= \frac{0.78}{\sqrt{0.666}}$$

$$= 0.03$$

Decision Trees (DT)

- DT : Construct a **tree where nodes correspond to tests on individual features**
  - ***can be useful*** for finding meaningful features
  - However, the feature set is very large, and we might find **spurious** correlations.

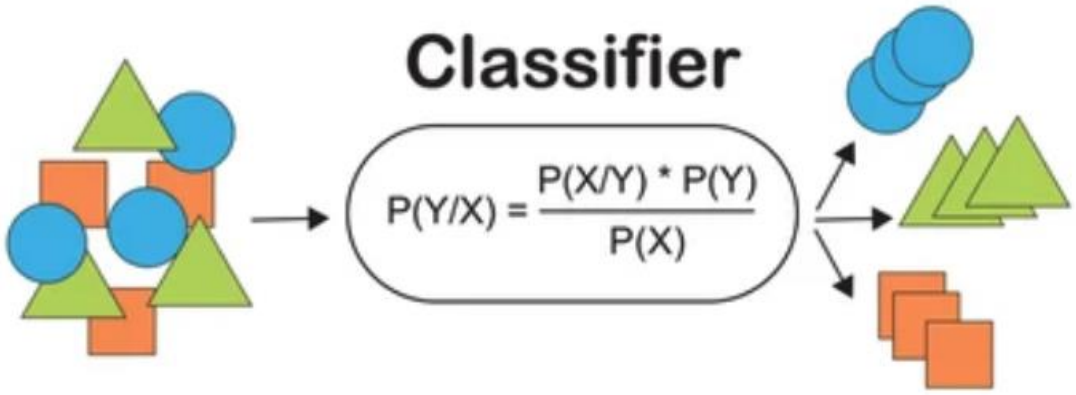
- Decision Trees using **Information Gain**
  - ***Information Gain :*** *to determine the best features for splitting nodes.*
  - ***poor choice*** because it tends to prefer **rare** features.



https://www.nltk.org/book/ch06.html

Naive Bayes (NB)

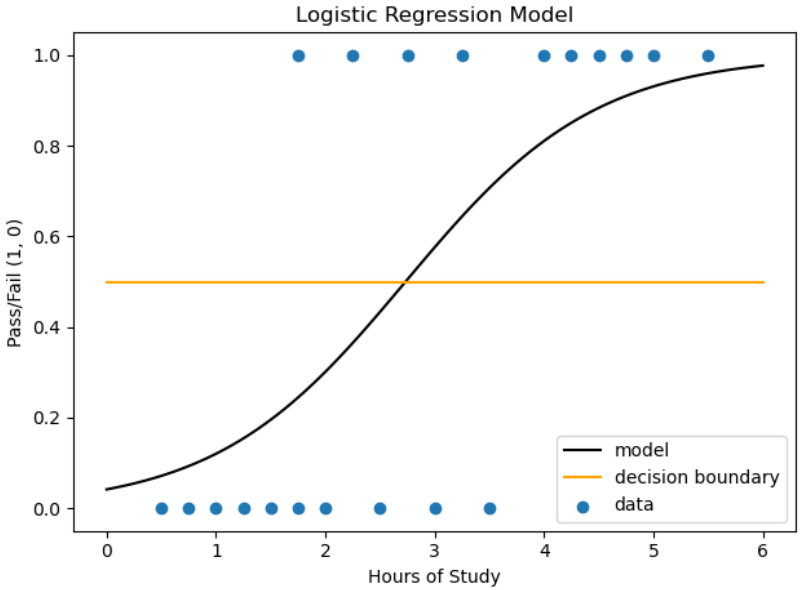
- NB : Find the class with **the highest likelihood under Bayes Law**
  - At first glance, a ***poor choice***
  - the "naive" assumption of the **conditional independence** of features and classes is highly untrue.
  - Also **sensitive to a large feature set**. Because many (small) probabilities are multiplied together, even uninformative features (irrelevant words) can lead to biased interpretations.
  - Surprisingly somewhat ***useful anyway***!



Logistic Regression (LR)

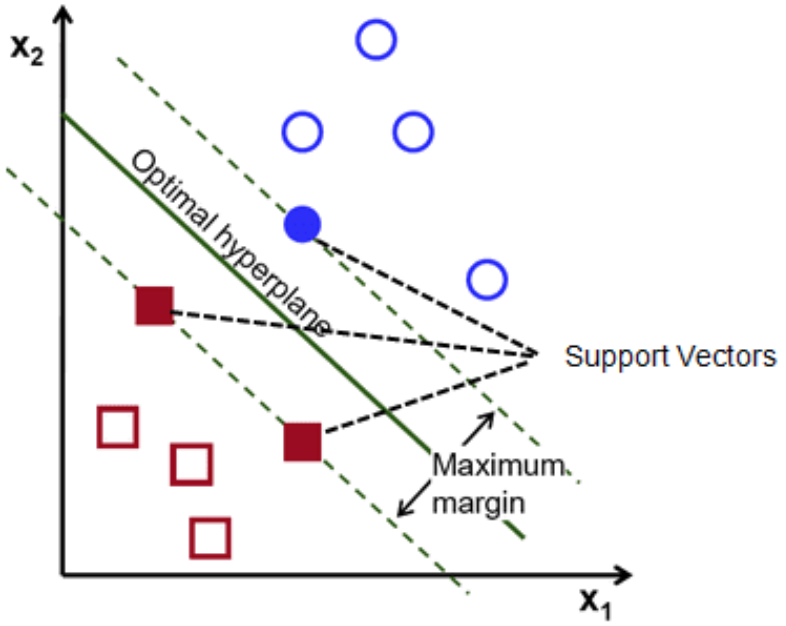
- LR : Put linear combination of features in **logistic function**
  - ***Useful***
  - it **relaxes the conditional independence requirement** of Naive Bayes.
  - Can handle large numbers of mostly useless features by **'feature weighting' step**

$$y = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$



Support Vector Machines (SVM)

- SVM : Finds **hyperplane** which **separates the training data** with **maximum margin**
  - Linear kernels often quite ***effective.***
  - Some combination of features are useful for characterising the classes.
  - Need substantial re-framing: **multiple classes** (most text classification tends to be multi–class).



Discussion : N-gram Language Model (LM)

2. For the following "corpus" of two documents:

```
1.  how much wood would a wood chuck chuck if a wood chuck
would chuck wood
2.  a wood chuck would chuck the wood he could chuck if
a wood chuck would chuck wood
```

(a) Which of the following sentences: `a wood could chuck`; `wood would a chuck`; is more probable, accoding to:

i. An unsmoothed uni-gram language model?
ii. A uni-gram language model, with Laplacian ("add-one") smoothing?
iii. An unsmoothed bi-gram language model?
iv. A bi-gram language model, with Laplacian smoothing?
v. An unsmoothed tri-gram language model?
vi. A tri-gram language model, with Laplacian smoothing?

	unsmooth	Laplacian
unigram	1)	2)
bi-gram	3)	4)
tri-gram	5)	6)

N-gram LM Calculation

- Corpus

1: how much wood would a wood chuck chuck if a wood chuck would chuck wood

2: a wood chuck would chuck the wood he could chuck if a wood chuck would chuck wood

- Word counts

<s>	a	chuck	could	he	how	if	much	the	wood	would	</s>	**Total**
2	4	9	1	1	1	2	1	1	8	4	2	**34**

- M: the number of all words - sum of all word frequency ***(M = 34)***
- V: the number of unique words - vocabulary size ***(V = 11)***

Why is the start symbol <s> left out?

- *used internally to set context but not included in the final n-gram probabilities*
- *because they are **artificially inserted** and **do not represent actual text content**.*
- *While the end symbol </s> is typically included because **</s> helps the model learn the likelihood of ending** after certain sequences of words.*

N-gram: 1) Unigram unsmooth

a	chuck	could	he	how	if	much	the	wood	would	</s>	**Total**
4	9	1	1	1	2	1	1	8	4	2	**34**

- M: the number of all words - sum of all word frequency ***(M = 34)***
- V: the number of unique words - vocabulary size ***(V = 11)***

- **Unigram probability (Unsmooth)**

$$P(w_i) = \frac{C(w_i)}{M}$$

M ← Total number of **word tokens** in corpus

A: a wood could chuck

B: wood would a chuck

$$\begin{aligned} P(A) &= P(\text{a})P(\text{wood})P(\text{could})P(\text{chuck})P(\text{</s>}) \\ &= \frac{4}{34} \times \frac{8}{34} \times \frac{1}{34} \times \frac{9}{34} \times \frac{2}{34} \approx 1.27 \times 10^{-5} \\ P(B) &= P(\text{wood})P(\text{would})P(\text{a})P(\text{chuck})P(\text{</s>}) \\ &= \frac{8}{34} \times \frac{4}{34} \times \frac{4}{34} \times \frac{9}{34} \times \frac{2}{34} \approx 5.07 \times 10^{-5} \end{aligned}$$

N-gram: 2) Unigram Laplacian smoothing

a	chuck	could	he	how	if	much	the	wood	would	</s>	**Total**
4	9	1	1	1	2	1	1	8	4	2	**34**

- M: the number of all words - sum of all word frequency ***(M = 34)***
- V: the number of unique words - vocabulary size ***(V = 11)***

- **Unigram Laplacian ("add-one") smoothing**

$$P_{add1}(w_i) = \frac{C(w_i) + 1}{M + |V|}$$

A: a wood could chuck

B: wood would a chuck

$$\begin{aligned} P_{\mathrm{L}}(A) &= P_{\mathrm{L}}(\mathtt{a})P_{\mathrm{L}}(\mathtt{wood})P_{\mathrm{L}}(\mathtt{could})P_{\mathrm{L}}(\mathtt{chuck})P_{\mathrm{L}}(\mathtt{</s>}) \\ &= \frac{5}{45} \times \frac{9}{45} \times \frac{2}{45} \times \frac{10}{45} \times \frac{3}{45} \approx 1.46 \times 10^{-5} \\ P_{\mathrm{L}}(B) &= P_{\mathrm{L}}(\mathtt{wood})P_{\mathrm{L}}(\mathtt{would})P_{\mathrm{L}}(\mathtt{a})P_{\mathrm{L}}(\mathtt{chuck})P_{\mathrm{L}}(\mathtt{</s>}) \\ &= \frac{9}{45} \times \frac{5}{45} \times \frac{5}{45} \times \frac{10}{45} \times \frac{3}{45} \approx 3.66 \times 10^{-5} \end{aligned}$$

N-gram: 3) bi-gram unsmooth

1: how much wood would a wood chuck chuck if a wood chuck would chuck wood

2: a wood chuck would chuck the wood he could chuck if a wood chuck would chuck wood

a	chuck	could	he	how	if	much	the	wood	would	</s>	**Total**
4	9	1	1	1	2	1	1	8	4	2	**34**

- **bi-gram probability (Unsmooth)**

$$P(w_i \mid w_{i-1}) = \frac{C(w_{i-1}w_i)}{C(w_{i-1})}$$

A: a wood could chuck

B: wood would a chuck

$$\begin{aligned} P(A) &= P(\text{a}|\text{<s>})P(\text{wood}|\text{a})P(\text{could}|\text{wood})P(\text{chuck}|\text{could})P(\text{</s>}|\text{chuck}) \\ &= \frac{1}{2} \times \frac{4}{4} \times \frac{0}{8} \times \frac{1}{1} \times \frac{0}{9} = 0 \\ P(B) &= P(\text{wood}|\text{<s>})P(\text{would}|\text{wood})P(\text{a}|\text{would})P(\text{chuck}|\text{a})P(\text{</s>}|\text{chuck}) \\ &= \frac{0}{2} \times \frac{1}{8} \times \frac{1}{4} \times \frac{0}{4} \times \frac{0}{9} = 0 \end{aligned}$$

N-gram: 4) bi-gram Laplacian smoothing

1: how much wood would a wood chuck chuck if a wood chuck would chuck wood

2: a wood chuck would chuck the wood he could chuck if a wood chuck would chuck wood

a	chuck	could	he	how	if	much	the	wood	would	</s>	**Total**
4	9	1	1	1	2	1	1	8	4	2	**34**

- M: the number of all words - sum of all word frequency ***(M = 34)***
- V: the number of unique words - vocabulary size ***(V = 11)***

- **bi-gram Laplacian ("add-one") smoothing**

$$P_{add1}(w_i \mid w_{i-1}) = \frac{C(w_{i-1}w_i) + 1}{C(w_{i-1}) + |V|}$$

A: a wood could chuck

B: wood would a chuck

$$P_L(A) = P_L(\text{a}|\text{<s>})P_L(\text{wood}|\text{a})P_L(\text{could}|\text{wood})P_L(\text{chuck}|\text{could})P_L(\text{</s>}|\text{chuck})$$
$$= \frac{2}{13} \times \frac{5}{15} \times \frac{1}{19} \times \frac{2}{12} \times \frac{1}{20} \approx 2.25 \times 10^{-5}$$

$$P_L(B) = P_L(\text{wood}|\text{<s>})P_L(\text{would}|\text{wood})P_L(\text{a}|\text{would})P_L(\text{chuck}|\text{a})P_L(\text{</s>}|\text{chuck})$$
$$= \frac{1}{13} \times \frac{2}{19} \times \frac{2}{15} \times \frac{1}{15} \times \frac{1}{20} \approx 3.60 \times 10^{-6}$$

N-gram: 5) tri-gram unsmooth

a	chuck	could	he	how	if	much	the	wood	would	</s>	**Total**
4	9	1	1	1	2	1	1	8	4	2	**34**

- M: the number of all words - sum of all word frequency ***(M = 34)***
- V: the number of unique words - vocabulary size ***(V = 11)***

- **tri-gram probability (Unsmooth)**

$$P(w_i|w_{i-1}w_{i-2}) = \frac{C(w_{i-2}w_{i-1}w_i)}{C(w_{i-2}w_{i-1})}$$

A: a wood could chuck

B: wood would a chuck

$$\begin{aligned} P(A) &= P(\text{a}|\text{<s> <s>})P(\text{wood}|\text{<s> a})\cdots P(\text{</s>}|\text{could chuck}) \\ &= \frac{1}{2}\times\frac{1}{1}\times\frac{0}{4}\times\frac{0}{0}\times\frac{0}{1} = ? \\ P(B) &= P(\text{wood}|\text{<s> <s>})P(\text{would}|\text{<s> wood})\cdots P(\text{</s>}|\text{a chuck}) \\ &= \frac{0}{2}\times\frac{0}{0}\times\frac{1}{1}\times\frac{0}{1}\times\frac{0}{0} = ? \end{aligned}$$

N-gram: 6) tri-gram Laplacian smoothing

a	chuck	could	he	how	if	much	the	wood	would	</s>	**Total**
4	9	1	1	1	2	1	1	8	4	2	**34**

- M: the number of all words - sum of all word frequency ***(M = 34)***
- V: the number of unique words - vocabulary size ***(V = 11)***

- **tri-gram Laplacian ("add-one") smoothing**

$$P_L(w_i|w_{i-1}w_{i-2}) = \frac{C(w_{i-2}w_{i-1}w_i)+1}{C(w_{i-2}w_{i-1})+V}$$

A: a wood could chuck

B: wood would a chuck

$$\begin{aligned} P_{\mathrm{L}}(A) &= P_{\mathrm{L}}(\texttt{a}|\texttt{<s> <s>})P_{\mathrm{L}}(\texttt{wood}|\texttt{<s> a})\cdots P_{\mathrm{L}}(\texttt{</s>}|\texttt{could chuck}) \\ &= \frac{2}{13}\times\frac{2}{12}\times\frac{1}{15}\times\frac{1}{11}\times\frac{1}{12}\approx 1.30\times10^{-5} \\ P_{\mathrm{L}}(B) &= P_{\mathrm{L}}(\texttt{wood}|\texttt{<s> <s>})P_{\mathrm{L}}(\texttt{would}|\texttt{<s> wood})\cdots P_{\mathrm{L}}(\texttt{</s>}|\texttt{a chuck}) \\ &= \frac{1}{13}\times\frac{1}{11}\times\frac{2}{12}\times\frac{1}{12}\times\frac{1}{11}\approx 8.83\times10^{-6} \end{aligned}$$

N-gram: Kneser-Ney (KN) smoothing

(b) Based on the "corpus", the vocabulary = {a, chuck, could, he, how, if, much, the, wood, would, </s> }, and the **continuation counts** of the following words are given as follows:

- a = 2
- could = 1
- he = 1
- how = 0
- if = 1
- much = 1
- the = 1
- would = 2
- </s> = 1

*number of unique words in the vocabulary which appears before a word **w***

- **chuck** = {wood, chuck, would, could} = 4
- **wood** = {much, a, chuck, the} = 4

i. What is the continuation probability of chuck and wood?

1: how much wood would a wood chuck chuck if a wood chuck would chuck wood
2: a wood chuck would chuck the wood he could chuck if a wood chuck would chuck wood

1: how much wood would a wood chuck chuck if a wood chuck would chuck wood
2: a wood chuck would chuck the wood he could chuck if a wood chuck would chuck wood

N-gram: Kneser-Ney (KN) smoothing

- **What is the continuation probability of chuck and wood?**

proportional to the number of observed contexts in which w_i appears

$$P_{cont}(w_i) = \frac{|\{w'_{i-1} : C(w'_{i-1}, w_i) > 0\}|}{\sum_{\{w_j : C(w_{i-1}, w_j) = 0\}} |\{w_{j-1} : C(w_{j-1}, w_j) > 0\}|}$$

Numerator: *# unique preceding words before* ***w***

Denominator: *Sum of all continuation count for total unique bigram types*

1: how much wood would a wood chuck chuck if a wood chuck would chuck wood

2: a wood chuck would chuck the wood he could chuck if a wood chuck would chuck wood

- a = {would, if} = **2**
- could = {he} = **1**
- he = {wood} = **1**
- how = { } = **0**
- if = {chuck} = **1**
- much = {how} = **1**
- the = {chuck} = **1**
- would = {wood, chuck} = **2**
- </s> = {wood} = **1**
- chuck = {wood, chuck, would, could} = **4**
- wood = {much, a, chuck, the} = **4**

18

$$P_{cont}(chuck) = \frac{\#_{cont}(chuck)}{18} = \frac{4}{18}$$

$$P_{cont}(wood) = \frac{\#_{cont}(wood)}{18} = \frac{4}{18}$$

N-gram: Recap

- bi-gram probability (Unsmooth)

$$P(w_i \mid w_{i-1}) = \frac{C(w_{i-1}w_i)}{C(w_{i-1})}$$

- bi-gram Laplacian ("add-one") smoothing

$$P_{add1}(w_i \mid w_{i-1}) = \frac{C(w_{i-1}w_i) + 1}{C(w_{i-1}) + |V|}$$

- bi-gram Kneser-Ney (KN) smoothing

$$P_{KN}(w_i \mid w_{i-1}) = \begin{cases} \frac{C(w_{i-1}, w_i) - D}{C(w_{i-1})}, & \text{if } C(w_{i-1}, w_i) > 0 \\ \beta(w_{i-1}) P_{cont}(w_i), & \text{otherwise} \end{cases}$$

the amount of probability mass that has been discounted for context w_{i-1}

$$P_{cont}(w_i) = \frac{|\{w'_{i-1} : C(w'_{i-1}, w_i) > 0\}|}{\sum_{\{w_j : C(w_{i-1}, w_j) = 0\}} |\{w_{j-1} : C(w_{j-1}, w_j) > 0\}|}$$

N-gram: back-off and interpolation

Back-off

- Use **lower-order n-gram model** if higher-order is unseen.
- For example, if we have never seen some tri-gram from our sentence, we can instead consider the bigram probability.

interpolation

- Take **weighted average sum of n-gram**.
- Instead of only “falling back” to lower (back-off), consider every probability as a linear combination of all of the relevant n-gram models.
- The weights are once more chosen to ensure that the probabilities of all events, given some context, sum to 1.

$$\begin{aligned} \mathrm{p}_{\text{Interpolation}}(w_m \mid w_{m-1}, w_{m-2}) = \lambda_3 \mathrm{p}_3^*(w_m \mid w_{m-1}, w_{m-2}) \\ + \lambda_2 \mathrm{p}_2^*(w_m \mid w_{m-1}) \\ + \lambda_1 \mathrm{p}_1^*(w_m). \end{aligned}$$

Programming!

Programming

1. In the `03-classification` notebook, observe how different tokenisation regimes alter the text classification performance of the various classifiers on the given Reuters dataset problem.

   (a) Alter the tokenisation strategy so that it incorporates other stages, for example, punctuation, or stemming/lemmatisation.

   (b) Does performance increase or decrease? Are some classifiers affected more than others? Why do you think that is?

2. Using the iPython notebook `04-ngram`, randomly generate some sentences based on the bi-gram models of the Gutenberg corpus and the Penn Treebank. What do you notice about these sentences? Are there any sentences which might get returned for both corpora? Why?

Programming!

3. Find a sentence with a higher probability than *revenue increased last quarter.*, according to:
   (a) The Gutenberg corpus, using bi-grams smoothed with Laplacian smoothing
   (b) The Gutenberg corpus, using bi-grams smoothed with Interpolation
   (c) The Penn Treebank corpus, using bi-grams and Laplacian smoothing
   (d) The Penn Treebank corpus, using bi-grams and Interpolation
4. Find the perplexity of the above (smoothed) language models for a number of sentences. Why does Interpolation generally have better perplexity?